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# Venn diagrams (v\11.12.3+)

Venn diagrams show relationships between sets using overlapping circles.

**Warning** This is a new diagram type in Mermaid. Its syntax may evolve in future versions.

## Syntax

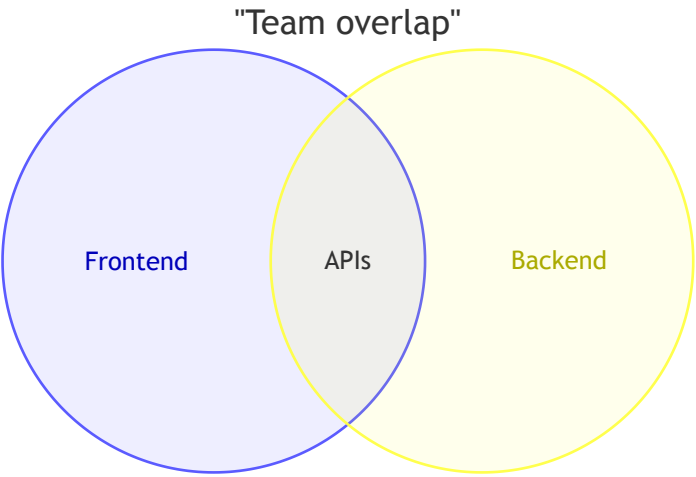
- Start with `venn-beta` .
- Use `set` for a single set name.
- Use `union` for an overlap of two or more set names.
- Identifiers in `union` must be defined by earlier `set` lines.
- Set identifiers can be bare words ( `A` , `Set_1` ) or quoted strings ( `"Foo Bar"` ).

Code:

```
venn-beta
  title "Team overlap"
  set Frontend
  set Backend
  union Frontend,Backend["APIs"]
```

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## Labels

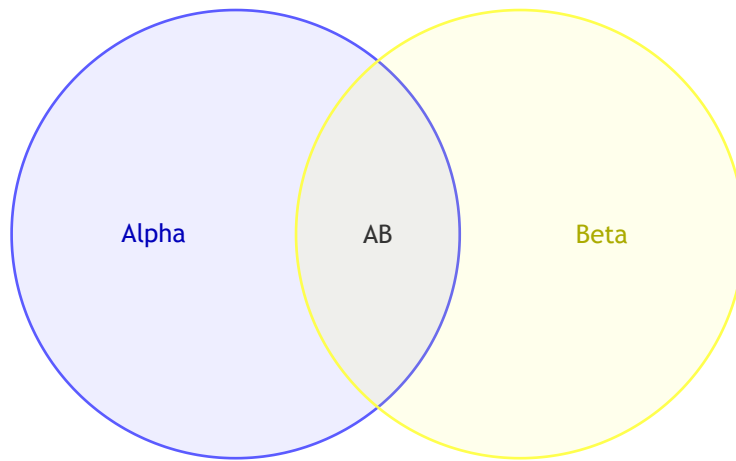
Use bracket syntax `["..."]` to set a display label while keeping the identifier short:

Code:

```
venn-beta
  set A["Alpha"]
  set B["Beta"]
  union A,B["AB"]
```

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## Sizes

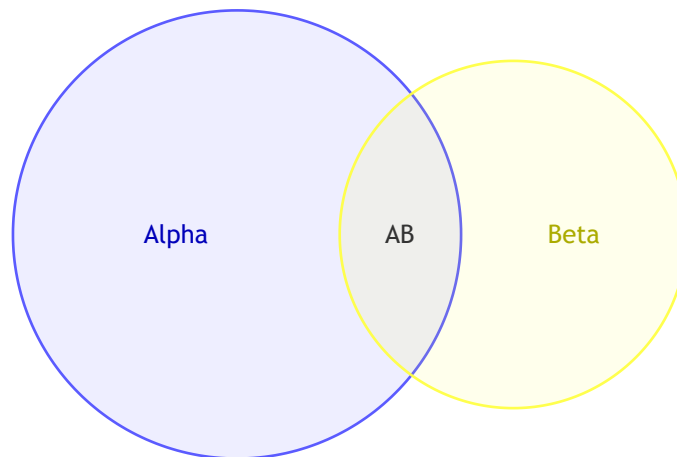
Use **:N** suffix to set the size of a set or union:

Code:

```
venn-beta
  set A["Alpha"]:20
  set B["Beta"]:12
  union A,B["AB"]:3
```

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## Text nodes

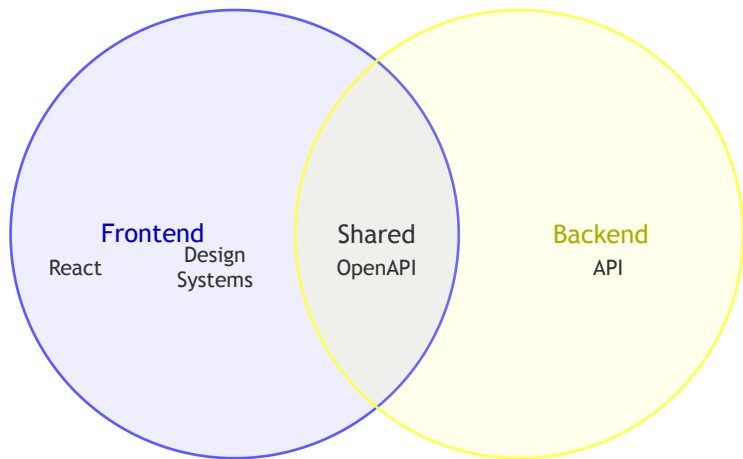
- Use **text** to place labels inside a set or union.
- Indented **text** lines attach to the most recent **set** or **union**.
- Use bracket syntax **["..."]** to set a display label for text nodes.

Code:

```
venn-beta
  set A["Frontend"]
    text A1["React"]
    text A2["Design Systems"]
  set B["Backend"]
    text B1["API"]
  union A,B["Shared"]
    text AB1["OpenAPI"]
```

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## Styling

Use `style` statements to apply visual styles to sets, unions, and text nodes:

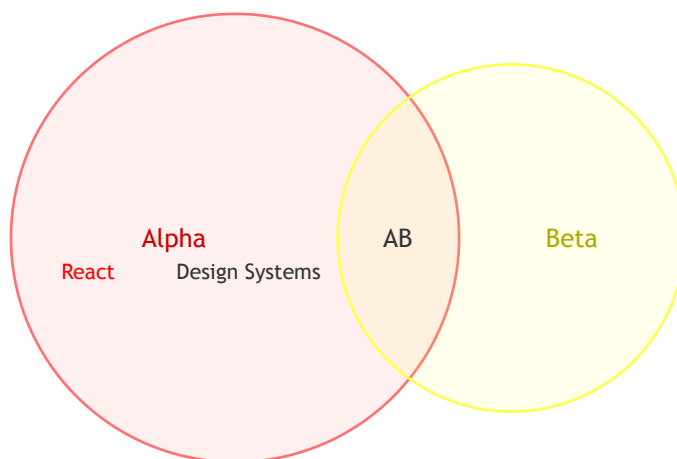
- `fill` : change the fill color
- `color` : change the text color
- `stroke` : change the stroke color
- `stroke-width` : change the stroke width
- `fill-opacity` : change the fill opacity

### Code:

```
venn-beta
  set A["Alpha"]:20
    text A1["React"]
    text A2["Design Systems"]
  set B["Beta"]:12
  union A,B["AB"]:3
  style A fill:#ff6b6b
  style A,B color:#333
  style A1 color:red
```

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